

EP2108 - Special Theory of Relativity

Assignment 1

Topics: Postulates of SR, Lorentz Transformations, Time Dilation, Length Contraction

Instructions: Show all derivations. Numerical answers without working will receive no credit.

Take $c = 3 \times 10^8$ m/s unless stated otherwise.

1. Foundations (Conceptual + Derivation)

- (a) State the two postulates of special relativity precisely. Explain why Galilean relativity is inconsistent with the constancy of the speed of light.
- (b) Starting *only* from the postulates (not by quoting the final formula), derive the Lorentz transformation equations for x', t' in terms of x, t for two frames S and S' moving with relative velocity v along the common x -axis. You may assume linearity of the transformation but must justify it.

2. Four-Vectors and Invariants — Basics

- (a) Define the position four-vector $x^\mu = (ct, x, y, z)$ and write down the Lorentz transformation matrix $\Lambda^\mu{}_\nu$ for a boost along x with velocity v . Verify explicitly that $\Lambda^T \eta \Lambda = \eta$.
- (b) Show that the four-velocity $u^\mu = \frac{dx^\mu}{d\tau}$ (where τ is proper time) satisfies $u^\mu u_\mu = c^2$ in *every* inertial frame, and explain why this is a stronger statement than merely being “the same number” — it must hold with the components evaluated in each frame separately.
- (c) Given a particle with four-momentum $p^\mu = (E/c, \vec{p})$, show that $p^\mu p_\mu = m^2 c^2$ is Lorentz invariant, and interpret this physically as the statement that rest mass is frame-independent.

3. Velocity and Acceleration Transformation

A particle moves with velocity components (u_x, u_y) in frame S . Frame S' moves with velocity v along $+x$ relative to S .

- (a) Derive the relativistic velocity transformation for u'_x and u'_y .

- (b) Show that if $u_x = u_y = c/\sqrt{2}$ in S , the speed of the particle in S is c , and verify explicitly that the transformed speed in S' is also exactly c , for arbitrary $v < c$.
- (c) Now derive the transformation law for acceleration a'_x in terms of u_x, a_x, a_y, v . Comment on why, unlike in Newtonian mechanics, force and acceleration are **not** parallel in general in relativistic dynamics.

4. The Twin Paradox — Quantitative Resolution

Twin A stays on Earth. Twin B travels to a star 12 light-years away (Earth frame) at speed $v = 0.8c$, decelerates instantaneously, and returns at the same speed.

- (a) Compute the total time elapsed for each twin.
- (b) Using the concept of *relativity of simultaneity* (not just time dilation), explain quantitatively how Twin B accounts for Twin A aging more, from B's own reference frame, during the outbound and inbound legs **and** during the turnaround. (Hint: analyze the sudden jump in "Earth's clock reading" as perceived in B's instantaneously changing rest frames at turnaround — the "simultaneity gap.")
- (c) Show that the paradox is resolved because B's worldline is not inertial throughout, and compute the magnitude of the simultaneity jump at turnaround explicitly.

5. Length Contraction and the Barn–Ladder Paradox

A ladder of proper length $L_0 = 10$ m moves horizontally at speed $v = 0.6c$ toward a barn of proper length $L_B = 6$ m, which has doors at both ends.

- (a) In the barn's rest frame, show that the ladder fits entirely inside the barn at some instant, and find the time interval during which both doors could be simultaneously closed.
- (b) In the ladder's rest frame, show explicitly using Lorentz transformations (not hand-waving) that the ladder does *not* fit inside the barn, and reconcile this with part (a) using relativity of simultaneity. You must show the front and back door closing *events* in both frames with coordinates (x, t) and (x', t') .
- (c) Suppose the far door is designed to open automatically exactly when the front of the ladder reaches it (in the barn frame), and the near door closes right after the back of the ladder enters. Determine whether the ladder ever suffers a "collision" in either frame's description, and argue why causality is preserved.

6. Minkowski Diagram and Invariant Interval

Two events occur: Event 1 at $(x_1, t_1) = (0, 0)$ and Event 2 at $(x_2, t_2) = (600 \text{ m}, 1 \text{ } \mu\text{s})$ in frame S .

- (a) Compute the spacetime interval $\Delta s^2 = c^2\Delta t^2 - \Delta x^2$. Determine whether the interval is timelike, spacelike, or null, and justify what this means physically for the possible causal relationship between the events.
- (b) Find the velocity v of a frame S' in which the two events are simultaneous, if such a frame exists. If it doesn't exist, explain why not.
- (c) Find the velocity of a frame in which the two events occur at the *same location*, if such a frame exists.

7. Cosmic-Ray Muons Reaching Earth's Surface

Muons are created high in the atmosphere at a height of $h = 3$ km, moving straight down toward the ground at speed $v = 0.98c$. In their own rest frame, muons live on average for $\tau_0 = 2 \mu\text{s}$ before decaying.

- (a) If we (wrongly) ignore relativity and just use $\tau_0 = 2 \mu\text{s}$ as the lifetime in *every* frame, calculate the distance a muon would travel before decaying. Compare this to $h = 3$ km — would the muon classically be expected to reach the ground?
- (b) Now include time dilation. Calculate γ for $v = 0.98c$, and find the dilated lifetime $\tau = \gamma\tau_0$ as measured on Earth. Recalculate the distance the muon travels in this time, and check whether it now reaches the ground.
- (c) In one or two sentences, explain in words why time dilation is needed to explain why so many muons are actually detected at sea level.

8. Frame S' moves at $0.70c$ along the $+x$ -direction relative to frame S . A particle in S' moves with speed $0.85c$ at an angle 50° with respect to the $+x'$ -axis.

- (a) Resolve the velocity into u'_x and u'_y .
- (b) Find u_x and u_y in frame S .
- (c) Determine the speed of the particle in S .
- (d) Calculate its direction of motion in S .

9. A spacecraft A moves at $0.65c$ relative to Earth. A smaller craft B is launched forward from A with speed $0.70c$ relative to A . A probe is then launched backward from B with speed $0.50c$ relative to B .

- (a) Find the velocity of B relative to Earth.
- (b) Find the velocity of the probe relative to Earth.
- (c) Find the velocity of the probe relative to spacecraft A .

10. A spacecraft has a proper length of 150 m and moves at $0.80c$ through a tunnel of proper length 120 m.

Two gates are installed at the entrance and exit of the tunnel.

- (a) Find the spacecraft length in the tunnel frame.
 - (b) Can both gates be closed simultaneously while the spacecraft is completely inside the tunnel?
 - (c) Calculate the tunnel length in the spacecraft frame.
 - (d) Suppose the two gates close simultaneously in the tunnel frame. Find the time separation between the two closing events in the spacecraft frame.
 - (e) Determine which gate closes first according to the spacecraft.
- 11.** Observers A and B are at rest relative to each other and separated by 6 light-years. Their clocks are synchronized in their common rest frame.
- Observer C passes A at $0.80c$ when all relevant local clocks read zero.
- (a) How long does C take to reach B according to A and B ?
 - (b) What time does C 's clock show when C reaches B ?
 - (c) In C 's frame, calculate the contracted separation between A and B .
 - (d) Calculate the reading of B 's clock that C assigns as simultaneous with the event of passing A .
- 12.** A spacecraft S' moves at $0.75c$ relative to Earth. At $t' = 4.0 \mu\text{s}$, a probe is launched from position $x' = 500 \text{ m}$ with velocity $0.60c$ in the $+x'$ -direction.
- (a) Transform the launch event into the Earth frame.
 - (b) Find the velocity of the probe relative to Earth.
 - (c) Write the equation of motion $x(t)$ of the probe in the Earth frame.
 - (d) Find the Earth-frame time when the probe reaches $x = 3000 \text{ m}$.